MoleHole: Real-time Black Hole Simulation using Compute Shaders

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Abstract

This paper presents the implementation details and theoretical background of MoleHole, a real-time black hole simulation application. By leveraging OpenGL 4.6 compute shaders, we achieve interactive frame rates for visualizing Schwarzschild and Kerr metrics. The system integrates a node-based animation system for dynamic scene control and utilizes a hybrid rendering approach combining rasterization for standard geometry and ray-marching for relativistic effects. We discuss the challenges of numerical integration near the event horizon and the techniques used to mitigate artifacts.

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1 Introduction

The visualization of General Relativity (GR) phenomena has traditionally been the domain of offline rendering due to the high computational cost of ray-tracing in curved spacetime. However, modern GPUs allow for real-time integration of geodesics. In this work, we explore the capabilities of current hardware to render rotating Black Holes (BHs) using the Kerr metric.

2 Background

2.1 General Relativity

The spacetime around a rotating black hole is described by the Kerr metric. The path of light rays (null geodesics) is determined by the geodesic equation:

$$\frac{d^2x^\mu}{d\lambda^2} + \Gamma^\mu_{\alpha\beta}\frac{dx^\alpha}{d\lambda}\frac{dx^\beta}{d\lambda} = 0 \tag{1}$$

where $\Gamma^\mu_{\alpha\beta}$ are the Christoffel symbols.

2.2 Ray Marching

To visualize the scene, we employ a ray-marching algorithm implemented in compute shaders.

3 Implementation

The application is built using C++23 and OpenGL 4.6. The architecture separates the simulation layer from the rendering layer.

3.1 Compute Shaders

The core integration is performed in GLSL compute shaders. We use a 4th-order Runge-Kutta (RK4) integrator for the geodesic equations.

3.2 Animation System

A node-based graph system allows users to animate parameters of the BH and other scene objects dynamically.

4 Results

We achieve performance exceeding 60 FPS on modern dedicated GPUs at 1080p resolution.

5 Conclusion

MoleHole demonstrates that high-fidelity relativistic rendering is feasible in real-time applications.

Acronyms

BH Black Hole. 1

References

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